NEIGHBOURHOOD EXTENSIONS COMPARED UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that the local cohomology of a complex with power-torsion cohomology computes the internal hom from it. Used inside the counted proof (source lines 1738-1748). Replaces the complex by its local cohomology on both sides.
- The compatibility of local cohomology with base change for finitely generated ideals. Invoked inside the counted proof (source lines 1749-1751). Identifies the two local cohomology objects.
- The earlier statement comparing the functors under base change. Invoked at the end of the counted proof (source lines 1752-1758). Supplies the quasi-isomorphism that finishes the argument.

Source reference. Original lemma lemma-torsion-change-rings: sources/stacks-dualizing/pinned-original-source.tex, source lines 1579–1579. Navigation pointer only: the counted text refers to the Stacks item lemma-torsion-change-rings, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-torsion-flat-change-rings: sources/stacks-dualizing/pinned-original-source.tex, source lines 1697–1697. Navigation pointer only: the counted text refers to the Stacks item lemma-torsion-flat-change-rings, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $A \to B$ be a flat ring map and let $I \subset A$ be a finitely generated ideal such that $A/I \to B/IB$ is an isomorphism. For $K \in D_{I^\infty\text{-}torsion}(A)$ and $L \in D(A)$ the map*

$$R\operatorname{Hom}_A(K, L) \longrightarrow R\operatorname{Hom}_B(K \otimes_A B, L \otimes_A B)$$

is a quasi-isomorphism. In particular, if M, N are A-modules and M is I-power torsion, then the canonical map

$$\operatorname{Ext}^i_A(M, N) \longrightarrow \operatorname{Ext}^i_B(M \otimes_A B, N \otimes_A B)$$

is an isomorphism for all i.

Proof. Let $Z = V(I) \subset \operatorname{Spec}(A)$ and $Y = V(IB) \subset \operatorname{Spec}(B)$. Since the cohomology modules of K are I power torsion, the canonical map $R\Gamma_Z(L) \to L$ induces an isomorphism

$$R\operatorname{Hom}_A(K, R\Gamma_Z(L)) \to R\operatorname{Hom}_A(K, L)$$

in $D(A)$. Similarly, the cohomology modules of $K \otimes_A B$ are IB power torsion and we have an isomorphism

$$R\operatorname{Hom}_B(K \otimes_A B, R\Gamma_Y(L \otimes_A B)) \to R\operatorname{Hom}_B(K \otimes_A B, L \otimes_A B)$$

in $D(B)$. By Lemma lemma-torsion-change-rings we have $R\Gamma_Z(L) \otimes_A B = R\Gamma_Y(L \otimes_A B)$. Hence it suffices to show that the map

$$R\operatorname{Hom}_A(K, R\Gamma_Z(L)) \to R\operatorname{Hom}_B(K \otimes_A B, R\Gamma_Z(L) \otimes_A B)$$

is a quasi-isomorphism. This follows from Lemma lemma-torsion-flat-change-rings. □